

Summary Report

Cross-Region EC2 Backup Replication Using AWS Backup

1. Steps to Configure Cross-Region Backup

Step 1: Launch and Prepare the EC2 Instance

A Linux-based **t3.micro** EC2 instance was launched in **us-east-1**. Instance ID: **i-04cfe745666331faf**. The instance was verified to be running and prepared to be protected using AWS Backup.

Step 2: Create the Primary Backup Vault

A primary backup vault named **EC2-Primary-Backup-Vault** was created in **us-east-1**. This vault stores the original EC2 recovery points.

Step 3: Create the Disaster Recovery Backup Vault

The region was changed to **us-west-1** and a second vault named **EC2-DR-Backup-Vault** was created. It acts as the destination for recovery points copied from the primary region.

Step 4: Create the AWS Backup Plan

A backup plan named **EC2-Cross-Region-Backup-Plan** was created in the primary region. The plan manages the backup schedule, vault, retention, and cross-region copy configuration.

Step 5: Configure the Daily Backup Rule

Configuration	Value
Frequency	Daily
Backup Vault	EC2-Primary-Backup-Vault
Time Zone	Asia/Calcutta (UTC+05:30)
Backup Time	05:40 PM
Retention	7 days
Start Within	1 hour
Complete Within	7 days

Step 6: Configure Cross-Region Copy

The cross-region copy was configured from **us-east-1** to **us-west-1**, from **EC2-Primary-Backup-Vault** to **EC2-DR-Backup-Vault**, with 7-day copy retention.

Step 7: Assign the EC2 Instance

The EC2 instance was assigned using **EC2-Backup-Assignment**. Resource type was EC2 and the IAM role used was **AWSBackupDefaultServiceRole**.

Step 8: Verify the Backup Configuration

The backup plan, rule, regions, vaults, EC2 resource, daily frequency, and 7-day retention settings were reviewed.

Step 9: Create an On-Demand Backup

An on-demand backup was initiated to test the configuration without waiting for the scheduled backup. The backup was stored in **EC2-Primary-Backup-Vault**.

Step 10: Monitor the Backup Job

The backup job was monitored under AWS Backup → Jobs → Backup jobs until its status became **Completed**.

Step 11: Verify the Primary Recovery Point

The primary vault was checked for a recovery point. An example recovery point was **image/ami-00222ba82661e72f7** with Completed status.

Step 12: Monitor the Cross-Region Copy Job

The copy job was monitored under AWS Backup → Jobs → Copy jobs until it reached **Completed**.

Step 13: Verify the Destination Recovery Point

The region was changed to **us-west-1** and **EC2-DR-Backup-Vault** was checked. An example replicated recovery point was **image/ami-06e2ca6440b6a7f4a** with Completed status.

2. Reason and Benefits of Cross-Region Backup Replication

1. **Disaster Recovery** – Provides a recovery option if the primary AWS Region becomes unavailable due to a major failure or disruption.
2. **Geographical Redundancy** – Stores a backup copy in a different AWS Region, reducing dependence on a single region.
3. **Protection Against Regional Failure** – The backup stored in the DR region provides additional protection if the primary region is affected.
4. **Data Protection** – Maintains an additional copy of the EC2 recovery point to protect against accidental data loss and infrastructure failures.
5. **Business Continuity** – Helps organizations recover critical infrastructure and continue operations after a major failure.
6. **Automated Backup and Replication** – AWS Backup automatically creates the scheduled backup and copies it to the DR region, reducing manual effort.
7. **Retention Management** – Recovery points can be automatically retained for a defined period. In this project, both backups and copies were configured with a **7-day retention period**.
8. **Centralized Backup Management** – AWS Backup provides a centralized interface for managing backup plans, rules, vaults, resource assignments, backup jobs, copy jobs, and recovery points.
9. **Improved Recovery Readiness** – Successfully maintaining recovery points in both **us-east-1** and **us-west-1** ensures that an additional recovery copy is available when required.

3. Issues Encountered and How They Were Resolved

Issue 1: Duplicate CopyActions Error

Problem:

While configuring the cross-region copy rule, the following error was encountered:

Duplicate CopyActions provided in backupRule.

This occurred because more than one identical cross-region copy action was configured in the backup rule.

Resolution:

The **Daily-EC2-Backup** rule was edited and the **Copy to destination** configuration was reviewed. Duplicate copy actions were removed, leaving only the required configuration:

- Destination Region: **us-west-1**
- Destination Vault: **EC2-DR-Backup-Vault**
- Copy Retention: **7 days**

The rule was saved again after removing the duplicate configuration.

Issue 2: Recovery Point Was Not Immediately Visible

Problem:

After initiating an on-demand backup, the recovery point was not immediately visible in the primary backup vault.

Resolution:

The backup job was monitored through:

AWS Backup → Jobs → Backup jobs

The job was allowed to complete processing. After the status changed to **Completed**, the **EC2-Primary-Backup-Vault** was checked again. The recovery point was then successfully displayed.

Issue 3: Cross-Region Recovery Point Was Not Immediately Visible

Problem:

After the primary backup completed, the replicated recovery point was not immediately visible in the destination backup vault.

Resolution:

The copy operation was monitored through:

AWS Backup → Jobs → Copy jobs

The copy job was allowed to complete. The region was then changed to **us-west-1**, and the **EC2-DR-Backup-Vault** was opened. The Recovery Points section was checked again, and the replicated recovery point was successfully found with a **Completed** status.

This confirmed that the delay was related to the time required for the cross-region copy operation to finish.